AlgoProfessor AI R&D Solutions

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From: Dr. S. Satyanarayana Ph.D., PDF(AI) — Founder & CEO ceo@algoprofessor.com

To: Sheshikala Mamidisetti mamidisettishashikala@gmail.com | +91 7893990205

Date: February 22, 2026 Ref: ALGO/INT/2026/009

Subject: 3-Month AI R&D Internship Offer — AI Foundations to Agentic AI Systems | AlgoProfessor Batch 2026 CC: AI Team Lead teamlead@algoprofessor.com

Attach: [1] Internship Offer Letter [2] 90-Day Work Schedule [3] GitHub Repository Guide

Dear Sheshikala,

Greetings from AlgoProfessor AI R&D Solutions! We are delighted to offer you a place in our flagship 3-Month Immersive Internship — AI Foundations to Agentic AI Systems, Batch 2026, commencing February 22, 2026.

Your profile is impressive and ideally matched for this programme: B.Tech ECE student (St. Martin's Engineering College, 3rd Year) with a strong academic foundation including a Diploma in ECE (CGPA: 9.1) and SSC (CGPA: 9.8), a comprehensive and self-driven ML skill set spanning Linear & Logistic Regression, Decision Tree, Random Forest, SVM, K-Means, PCA, LDA, and SVD, solid Python, SQL, NumPy, Pandas, Matplotlib, Seaborn, Power BI data stack, and — most notably — an active IIT Indore Drishti CPS Foundation Professional Certification Program in AI & Data Science (expected 09/2027). Your structured academic approach to ML theory combined with hands on data science tools makes you exceptionally well-prepared to accelerate through this programme.

- Attachment 1 — Offer Letter: Role, responsibilities, benefits, grading, and terms & conditions. • Attachment 2 — 90-Day Schedule: Phase-by-phase week-by-week plan across Phase 1 (Foundations), Phase 2 (LLM Engineering), and Phase 3 (Agentic AI + Grand Capstone: DataSense AI).
- Attachment 3 — GitHub Guide: Repository structure, daily commit conventions, and reference repo (https://github.com/228W1A1274/algoprofessor-ai-internship).

Internship Snapshot

Programme	AI → Agentic AI — Batch 2026
Duration	90 Days (Feb 22 – May 22, 2026)
Projects	16+ Milestones + Grand Capstone
GitHub	github.com/sheshikala-m/algoprofessor-ai-internship — Daily by 5 PM
LinkedIn	linkedin.com/in/sheshikala-m-b4a6a332b
Mode	Remote / Hybrid
Stipend	As per company policy
Phase 1 (D1–30)	Git, Python, ML, DL, RAG — 5 milestones
Phase 2 (D31–60)	LLM Engineering, Fine-tuning — 4 milestones
Phase 3 (D61–90)	Agentic AI, MCP, Deployment — 7 milestones
Grand Capstone (D90)	DataSense AI — 4 agents + 40+ tools

Please sign and return the Offer Letter by February 27, 2026. Queries: ceo@algoprofessor.com. We look forward to welcoming you to the AlgoProfessor AI family!

Dr. S. Satyanarayana Ph.D., PDF(AI) | Founder & CEO, AlgoProfessor AI R&D Solutions | ceo@algoprofessor.com

Acceptance of Offer: Sheshikala Mamidisetti(Sheshikala **) Date:22-02-2026**

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ATTACHMENT 1 — INTERNSHIP OFFER LETTER

Ref: ALGO/INT/2026/009 | Date: February 22, 2026

To: Sheshikala Mamidisetti	Email: mamidisettishashikala@gmail.com
Qual.: B.Tech ECE 3rd Year + IIT Indore AI/DS Cert (Pursuing)	Phone: +91 7893990205

Subject: Internship Offer — 3-Month AI R&D Programme: AI Foundations to Agentic AI Systems, Batch 2026

Dear Sheshikala, we are delighted to offer you a place in the AlgoProfessor 3-Month Internship Batch 2026.

Your strong theoretical grounding in the full ML algorithm stack (Regression, Decision Tree, Random Forest, SVM, K-Means, PCA, LDA, SVD), practical Python and SQL data analysis skills, visualization expertise with Matplotlib, Seaborn, and Power BI, and your active IIT Indore Drishti CPS AI & Data Science certification programme makes you an excellent fit. Over 90 days you will build 16+ production-grade AI projects and deliver the DataSense AI grand capstone — an autonomous intelligent data analysis and insights generation platform that directly amplifies your ML algorithms, statistical mathematics, and data science foundation into a full multi agent Agentic AI system.

Position Details

Programme	AI to Agentic AI — Batch 2026
Role	AI R&D Intern (Data Science & ML Track)
Duration	90 Days / 3 Months
Start	February 22, 2026
End	May 22, 2026
Milestones	16 Projects + Grand Capstone
GitHub	github.com/sheshikala-m/algoprofessor-ai-internship — Daily commit by 5 PM
Certification	IIT Indore Drishti CPS AI & Data Science (Pursuing — 09/2027)

Mode	Remote / Hybrid
Stipend	As per company policy

Key Responsibilities

- Daily production-grade code + GitHub commits (day-by-day folder structure)
- Build 16 milestone AI/Agentic AI projects leveraging full ML algorithm expertise
- Apply Python, NumPy, Pandas, SQL, and statistical skills to AI pipelines and data analysis tasks • Fine-tune LLMs (LoRA/QLoRA) on structured and time-series domain data
- Build multi-agent systems: LangGraph, CrewAI — applied to data intelligence workflows • Develop and deploy custom MCP servers including data analysis connectors
- Deliver Grand Capstone: DataSense AI
- Mentor reviews every 10 days; 3 phase reviews

What You Will Gain

- End-to-end Agentic AI build experience that complements your IIT Indore AI & DS certification • Mentorship from Dr. Satyanarayana + AI team
- 16+ industry-grade project portfolio on public GitHub
- Certificate of Excellence / Completion
- Letter of Recommendation
- LinkedIn Endorsement (top performers)
- Full-time employment opportunity — Data Science / AI track

Terms and Conditions

- Daily GitHub commit by 5 PM — no exceptions; folder per day, PEP 8 code quality • Complete all 16 milestones + grand capstone
- Maintain strict confidentiality of all IP
- Documentation standards: README per folder + root README weekly update • Periodic mentor evaluations every 10 days
- Score <60 may result in programme review

Grading: 90–100 = Certificate of Excellence + LOR + LinkedIn Endorsement | 75–89 = Completion Certificate + Endorsement | 60–74 = Participation Certificate | <60 = Mandatory Review

Please confirm acceptance by February 27, 2026 at ceo@algoprofessor.com.

Sincerely,

Dr. S. Satyanarayana Ph.D., PDF(AI)
Founder & Chief Executive Officer
AlgoProfessor AI R&D Solutions | ceo@algoprofessor.com

Candidate Acceptance: Sheshikala Mamidisetti(Sheshikala **) Date:22-02-2026**

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ATTACHMENT 2 — 90-DAY WORK SCHEDULE: AI TO AGENTIC AI

Intern: Sheshikala Mamidisetti | Feb 22 – May 22, 2026 | Ref: ALGO/INT/2026/009

Phase 1 (Days 1–30): Foundations — Python, ML, Deep Learning, Databases, RAG | 5 Milestones | 3 Mentor Reviews			
Wk	**Days/Dates**	**Key Tasks & Deliverables — Tailored to Sheshikala's Python/ML/Stats/Data Science Background**	**Milestone**
1	D1–5 Feb 22–28	day01_python_ai_fundamentals/ — Git/GitHub setup, Python 3.10+ advanced syntax, NumPy/Pandas EDA (directly extends existing NumPy/Pandas/EDA skills), Matplotlib/Seaborn dashboards (leverages existing visualization stack), OOP design patterns, Docker, Poetry project environments	**GitHub repo + EDA pipeline**
2	D6–10 Mar 1–7	day02_machine_learning/ + day03_feature_engineering/ — Advanced Scikit-learn pipelines (extends existing Linear/Logistic Regression, Decision Tree, Random Forest, SVM, K-Means, PCA, LDA, SVD knowledge into production-grade pipelines), XGBoost/LightGBM ensembles, PyTorch NN from scratch using Linear Algebra foundations (extends Statistics & Math skills)	**M1: Web Intelligence Synthesiser**
3	D11–15 Mar 8– 14	day_4_neuralnetworks/ + day11_databases/ — Deep neural networks with statistical theory grounding (extends Descriptive & Inferential Statistics, Linear Algebra), PostgreSQL + async SQLAlchemy (extends SQL skills), MongoDB + Redis caching, vector DBs (Chroma, Pinecone, FAISS), RAG architecture, Graph RAG with Neo4j — applied to data science datasets	**M2: Enterprise Knowledge Navigator**
4	D16–25 Mar 15– 24	day5_nlp/ + Day_6_Healthcare_Bot adapted for data analytics — OpenAI + Claude APIs, CoT/ReAct/DSPy prompt engineering, function calling + Pydantic structured outputs for data analysis reports, conversational memory, Llama 3/Mistral via Ollama — applied to structured data and time series contexts (complements Time Series learning)	**M3: DataAssist Analytics Agent**
5	D26–30 Mar 25– 31	Whisper audio AI for data briefings, AI-assisted data report generation (leverages IIT Indore AI & DS curriculum context), FastAPI + Docker REST analytics services, Phase 1 review sprint (pytest + mkdocs), LLM evaluation + NeMo guardrails	**M4+M5: InsightScribe + CodeXcelerate**
Phase 2 (Days 31–60): LLM Engineering — Frontier Models, Fine-tuning (LoRA/QLoRA), Evaluation | 4 Milestones | 3 Reviews			
Wk	**Days/Dates**	**Key Tasks & Deliverables — Tailored to Sheshikala's Python/ML/Stats/Data Science Background**	**Milestone**
6	D31–35 Apr 1–7	day8_agent/ style — LLM benchmarking (GPT-4o, Claude, Gemini, Llama) on data science benchmarks, Tree-of-Thought/DSPy reasoning (applies Inferential Statistics thinking), Pydantic AI structured outputs for automated ML model reports — directly aligns with IIT Indore AI & DS programme curriculum, Google Colab + Jupyter Notebook workflows (extends existing tools)	**M6: DataOracle — LLM insights predictor**

		— run in parallel with IIT Indore certification coursework	
7	D36–40 Apr 8– 14	day13/ — LoRA/QLoRA fine-tuning, TRL SFTTrainer on tabular/time series datasets (directly extends Time Series learning + full ML algorithm stack), DPO preference tuning for data analyst behaviour, GPTQ/AWQ quantisation, vLLM inference, W&B experiment tracking — complements IIT Indore certification modules	**M7: TimeSeriesHunter — QLoRA Llama 3.1**
8	D41–45 Apr 15– 21	day7_rag/ extended — RAG vs fine-tuning for data science knowledge bases, GPT-4V/LLaVA multi-modal (chart and dashboard image analysis — extends Power BI and Matplotlib/Seaborn skills), pgvector embeddings for statistical reports, MT-Bench/HELM eval, LangSmith + Arize Phoenix observability, Power BI integration (leverages existing BI skills)	**M8: InsightBot — Multi agent data analyst**
9	D46–60 Apr 22– May 6	day9_multiagent/ extended — NeMo Guardrails + Presidio PII redaction for sensitive data pipelines, FastAPI SSE streaming for real-time data feeds, AWS SageMaker + Azure ML deployment, Phase 2 review sprint — all aligned with IIT Indore AI & DS programme advanced modules	**Phase 2 Capstone + Cloud deploy demo**

Phase 3 (Days 61–90): Agentic AI — Multi-Agent Systems, MCP Servers, LangGraph, Grand Capstone | 7 Milestones | DataSense AI			
Wk	**Days/Dates**	**Key Tasks & Deliverables — Tailored to Sheshikala's Python/ML/Stats/Data Science Background**	**Milestone**
10	D61–65 May 7– 11	day14/ + beyond — Agent foundations (ReAct, Reflexion, Plan Execute), custom MCP servers (SQL data connector — extends SQL skills, REST analytics API, Power BI report server — extends BI knowledge, statistical model store with PCA/SVM/RF endpoints), LangGraph state machines + human-in-the-loop for data QA validation	**M9: TalentTwin — AI Career Digital Twin**
11	D66–70 May 12–16	Crew AI data science team (DataPlanner + StatAnalyst + MLEngineer + ReportWriter), automated EDA pipeline agent (directly extends NumPy/Pandas/EDA/Seaborn stack), K-Means clustering orchestrator + PCA dimensionality reduction pipeline as agentic tools — transforms existing ML algorithms into multi-agent workflows	**M10: DataSquad — 4-Agent Analytics Team**
12	D71–75 May 17–21	AutoGen conversable data agents + analytics group chat, meta agent builder for adaptive analysis strategies, SidekickBrowser (automated data research via LangGraph + Playwright), A2A multi agent comms via RabbitMQ/Redis Streams, time series forecasting agent pipeline (directly extends Time Series learning interest)	**M11: SidekickBrowser Data Co-pilot**
GC	D81–90 May 12–22	DataSense AI: Autonomous Intelligent Data Analysis & Insights Platform — 4 agents (DataAnalyst, StatisticsEngine, MLOrchestrator, InsightReporter) + 6 MCP servers (SQL data connector, Power BI report server, statistical model store with RF/SVM/PCA endpoints, time series forecasting engine, EDA pipeline server, insight knowledge base) + 40+ tools. Directly integrates Sheshikala's full ML algorithm stack (Random Forest, SVM, K-Means, PCA, LDA, SVD), statistical mathematics knowledge, SQL + Python data analysis skills, and Power BI visualization expertise. Docker/K8s + Grafana monitoring + 20-min live CEO panel demo.	**GRAND CAPSTONE: DataSense AI**

Duration	90 Days
Projects	16+ Milestones
Evaluations	9 (every 10 days)
GitHub	github.com/sheshikala-m/algoprofessor-ai-internship — Daily by 5:00 PM
Top Outcome	Certificate of Excellence + LOR

Approved by:

Dr. S. Satyanarayana Ph.D., PDF(AI)
Founder & CEO, AlgoProfessor AI R&D Solutions Signature & Date

ATTACHMENT 3 — GITHUB REPOSITORY GUIDE & DAILY COMMIT CONVENTION

Intern: Sheshikala Mamidisetti | Ref: ALGO/INT/2026/009

GitHub Repository Setup

Internship Repo to Create:
https://github.com/sheshikala-m/algoprofessor-ai-internship **LinkedIn Profile:**
https://linkedin.com/in/sheshikala-m-b4a6a332b
Reference Repo (Batch 2026 Example): https://github.com/228W1A1274/algoprofessor-ai-internship

Create a new public GitHub account with username sheshikala-m (or use your existing account). Study the reference repository's folder structure, README style, and commit history before Day 1. Your repo must follow the exact naming conventions shown below:

Required Folder Structure

Folder / File	Topic	Phase
`day01_python_ai_fundamentals/`	Python 3.10+, NumPy, Pandas, EDA, Git setup	**Phase 1**
`day02_machine_learning/`	Scikit-learn — RF, SVM, Decision Tree pipelines	**Phase 1**
`day03_feature_engineering/`	PCA, LDA, SVD, feature selection, preprocessing	**Phase 1**
`day_4_neuralnetworks/`	PyTorch NN using Linear Algebra foundations	**Phase 1**

`day5_nlp/`	HuggingFace Transformers, NLP for data reports	**Phase 1**
`Day_6_Healthcare_Bot/`	OpenAI/Claude API, data analytics chatbot	**Phase 1**
`day7_rag/`	RAG on statistics/ML knowledge bases, FAISS	**Phase 1**
`day8_agent/`	ReAct data analysis agent, LangChain, tool use	**Phase 2**
`day9_multiagent/`	Multi-agent EDA systems, Crew AI	**Phase 2**
`day10_project/`	Milestone project — DataOracle capstone	**Phase 2**
`day11_databases/`	PostgreSQL, SQLAlchemy, pgvector (extends SQL)	**Phase 2**
`day_12_mongodb_redis/`	MongoDB data logs, Redis analytics caching	**Phase 2**
`day13/`	LoRA/QLoRA fine-tuning on tabular/time series data	**Phase 2**
`day14/`	LangGraph data flows, MCP SQL/BI servers	**Phase 3**
`dayNN_.../ (continues daily)`	Agentic AI, cloud deploy, DataSense capstone	**Phase 3**
`.gitignore`	Standard Python .gitignore	**Root**
`README.md`	Project overview, ML algorithm log, IIT cert link	**Root**
`3months_Internship_Plan.pdf`	Signed internship plan (scanned copy)	**Root**

Daily Commit Rules

- Create a new day folder each working day: dayNN_topic_name/ (e.g. day15_timeseries_forecasting_agent/) • Each folder must contain: main script (.py or .ipynb via Google Colab/Jupyter), README.md with task description and results, and output files, charts, or screenshots
- Commit message format: Day NN: <Topic> — <Brief description> (e.g. Day 15: Time Series Agent — Built ARIMA + LLM hybrid forecasting agent with 91% accuracy)
- All commits must be pushed by 5:00 PM IST daily — late commits are marked as missed • Repository must remain PUBLIC throughout the internship duration
- README.md at root must be updated weekly with progress summary, completed milestones, and IIT Indore certification module alignment notes
- Languages expected: Python 71.8%, Jupyter Notebook 23.9%, HTML 2.5%, PLpgSQL/SQL 1.5%

README.md Template (Root Level)

```
# algoprofessor-ai-internship

AI R&D Internship — AlgoProfessor AI Software Solutions (Feb-May 2026)
Intern: Sheshikala Mamidisetti | Track: Data Science & ML Agentic AI
IIT Indore Drishti CPS AI & Data Science Certification (Pursuing)

## ML Algorithm Coverage
Linear/Logistic Regression | Decision Tree | Random Forest | SVM
K-Means | PCA | LDA | SVD | XGBoost | LightGBM | PyTorch DL

## Progress Tracker
- [x] Phase 1: Python, ML, DL, RAG, Databases
- [ ] Phase 2: LLM Engineering, Fine-tuning on Tabular/Time Series Data
- [ ] Phase 3: Agentic AI + Grand Capstone: DataSense AI

## Grand Capstone
DataSense AI — Autonomous Intelligent Data Analysis & Insights Platform
4 Agents | 6 MCP Servers | 40+ Tools | SQL + Power BI + ML Fusion
```